Razib Hasan

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ROBOTICS SOFTWARE ENGINEER

Bridging modern AI/LLMs with physical hardware control and real-time data engineering. Working with programming quadruped/humanoid robots (Unitree G1 Edu) and industrial manipulators (Omron), while architecting end-to-end MQTT telemetry pipelines for automated 3D printer networks. Applies clean architecture principles, robust testing, and containerized workflows to build scalable, maintainable robotics and edge-computing platforms.

TECHNICAL SKILLS

Microelectronics & Firmware

- **Circuit Design & Simulation:** Analog/Digital circuit design, PCB layout, schematic capture, System-on-Chip (SoC) fundamentals, LTspice, MATLAB, Mathematica.
- **Firmware & Hardware:** Low level development (C/C++), FreeRTOS, microcontrollers, bare-metal programming.
- **Bus & Hardware Protocols:** SPI, I2C, UART, CAN bus, TCP/IP, real-time sensor interfacing and signal acquisition.

Software & Data Engineering

- **Algorithms & Data Structures:** Algorithm design,, dynamic programming, computational complexity (C++, Python).
- **Data Engineering & Cloud:** Real-time data pipelines, Python data stack (Pandas, NumPy), SQL & schema design, Fiware, AWS (EC2, S3), Docker, Kubernetes.
- **Software & MLOps:** Clean code architecture, test-driven development (pytest), CI/CD, PyTorch, TensorFlow, Scikit-learn, MLOps deployment workflows.

Robotics & Nanotechnology

- **Robotics & Autonomous Systems:** Legged (Unitree G1) robotics engineering, ROS/ROS2 control software, real-time fleet telemetry, kinematics, sensor fusion.
- **Optics & Sensor R&D:** Near-Infrared (NIR) spectroscopy, optical system integration, Digital Signal Processing .
- **Nanotechnology & Physics:** Nanoelectronics, semiconductor physics, microelectronics design.

PROFESSIONAL EXPERIENCE

Tampere University of Applied Sciences
Automation Software Engineer (Fixed Term) **Jan 2026 - Dec 2026**

Architected end-to-end data ingestion and control infrastructure for heterogeneous robotic fleets (Unitree G1 Edu humanoids, Omron arms, and 3D printer networks). Integrated and fine-tuned LLMs on humanoid platforms using NVIDIA edge compute and GPU acceleration to enable real-time Embodied AI and task execution. Built Python ETL workflows and real-time MQTT streaming pipelines to aggregate, clean, and transform high-volume telemetry for performance analytics via Fiware. Established automated data validation (pytest) across distributed nodes while optimizing data throughput and latency for resource-constrained edge environments.

Solita Oy
Software Architect Intern **Jan 2024 - May 2024**

Designed and developed a mobile edge application for real-time event detection and geolocation tracking. Built backend services and APIs for scalable data ingestion and analytics, and created dashboards for monitoring and reporting. Worked in an agile team environment following modern software engineering practices.

EDUCATION

Aalto University
Master's Programme Electronics & Nanotech
Majors: Microelectronic circuit Design
GPA: 5.00/ 5.00 (ongoing)
2026-2028

University of Helsinki
Master's Programme in Computer Sciences
Majors: Algorithm
GPA: 4.82/ 5.00 (Studies done, only thesis ongoing)
2025-2027

Tampere University of Applied Sciences
Bachelor of Engineering
Majors: Software Engineering
GPA: 4.50/ 5.00
2021-2025

SELECTED PROJECTS

Autonomous Vehicle | 3D Printers Dashboard | Interview Challenge (Pytest)

ACHIEVEMENTS

Evertiq Scholarship, 2026	Position: 1st (Champion)
Jyväskylä Business, 2026	Audience award
Jyväskylä Business, 2026	Coup de cœur award

REFERENCE

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WORK RIGHT

Permit Type: P
Hold a Permanent Resident Permit (Finland)